

Kshitija Kumbharkar

Product Manager & Product Design Engineer

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Product and design engineer bridging hardware, software, and product strategy — from 0 → 1 ADAS platforms and EV systems integration to shipped GenAI applications. I turn engineering and stakeholder requirements into products people actually use.

EDUCATION

University of Southern California — USC Viterbi, M.S. Engineering Management	May 2026 · GPA 3.85
Vishwakarma Institute of Technology, Pune, B.Tech Mechanical Engineering	Jul 2022 · GPA 8.98/10

EXPERIENCE

Product Intern — Product Strategy & Platform Roadmap , <i>Porsche Engineering Services (PEUS)</i>	Jun 2025 – May 2026
- Led 0 → 1 development of Houston, an internal ADAS simulation platform, translating engineering and stakeholder requirements into a prioritized roadmap; scaled adoption from 10 to 50+ vehicles over a 12-month build. - Built and launched Houston_CC, a geospatial module tracking 50,000+ km of ADAS fleet coverage; developed night-driving and lane-change analytics that cut processing time from 4–5 hours to ~10 minutes. - Managed A2D2 coverage KPIs across two program phases and built automation to improve engineering workflows and prioritize technical work.	
Product Design Engineer — Systems Integration & PLM , <i>Mercedes-Benz Research & Development India</i>	Sep 2022 – Jul 2024
- Led wiring-harness design for an EV supplier program across the product lifecycle, translating requirements into release-ready designs using UG-NX, PREEVISION, and EPDM. - Improved UG-NX design-data accuracy by 20% by identifying workflow inconsistencies and refining specifications with engineering and supply-chain teams. - Supported PPAP risk assessments and ISO workflow development, driving 30% lower operating costs, 15% higher process efficiency, and ~2% S-Class cost savings.	
Design Intern — Operations & Stakeholder Management , <i>Dassault Systèmes</i>	Jan 2021 – May 2021
- Coordinated biweekly meetings across manufacturing, quality, and IT teams supporting the 3DEXPERIENCE platform, converting project issues and KPIs into actionable follow-ups. - Improved project delivery time by 20% and client satisfaction by 15% through structured stakeholder coordination and KPI tracking.	
Supply Chain Intern — Process Automation & Workflow Design , <i>Cummins</i>	Jun 2019 – Sep 2019
- Designed automated NHNT engine assembly-line solutions with two-hand safety controls and min-max inventory management, improving operational efficiency by 25%. - Analyzed inventory patterns across facilities and tuned replenishment parameters, reducing stockouts/overstock by 20% and holding costs by 10%.	

PROJECTS

The Daily Brief — Multi-Model GenAI News Summarization: Deployed a 0 → 1 GenAI app using four LLMs, Flask, and Hugging Face Spaces across 120 articles, with a ROUGE-1 / BERTScore + human evaluation framework comparing model quality. FSAE Race Car Suspension & Steering (Dassault): Designed a complete double-wishbone suspension and rack-and-pinion steering system for a Formula Student car in CATIA / 3DEXPERIENCE, iterating geometry in Catia Dymola. Kaplan Turbine Design & CFD: Designed a ~1 MW low-head hydro turbine with full SolidWorks CAD, CFD flow simulation, and static-structural analysis of the runner and casing.	
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TECHNICAL SKILLS

Engineering:	SolidWorks, UG-NX, PREEVISION, EPDM, CATIA V5, CREO, ANSYS, Motor-CAD, PLM, tolerance stack-up, wiring harness design
Software & Data:	Python, JavaScript, Node.js, SQL, Flask, GenAI/LLMs, geospatial systems, data analytics
Product & Program:	Product development, requirements, roadmap planning, milestones, dependencies, risk analysis, KPI development, stakeholder management, process improvement, Jira, Confluence, Figma

LEADERSHIP

Viterbi Graduate Students Association — Senator, ISE Department (Aug 2025 – May 2026): represented 150+ ISE students and organized industry events featuring senior leaders from Qualcomm, Western Digital, and FIS.
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